DIGITAL SAT MATH MASTERY SERIES

CREATING LINEAR MODELS FROM WORD PROBLEMS

Total Questions: 50 | Easy: 8 | Medium: 20 | Hard: 22

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Aligned to Digital SAT Question Style & Cognitive Demand

All questions original — fresh numbers, fresh contexts

Topic	Creating Linear Models from Word Problems
Document	DOCUMENT 2 — SOLUTIONS & EXPLANATIONS
Questions	50
Difficulty	Easy: 8 | Medium: 20 | Hard: 22
Format	Multiple Choice (4 options)
Exam Target	Digital SAT (DSAT)

■■■ CREATING LINEAR MODELS FROM WORD PROBLEMS — FULL SOLUTIONS & EXPLANATIONS ■■■

Q1. [EASY] **✓ Correct Answer: C) C = 1.75m + 3.50**

A taxi company charges a flat fee of $3.50 plus $1.75 per mile driven. Which of the following equations represents the total cost C, in dollars, for a ride of m...

STEP-BY-STEP SOLUTION:

Start with the flat fee of $3.50 (this is the y-intercept / fixed cost). Add the variable charge of $1.75 per mile: C = 1.75m + 3.50.

WHY EACH OPTION IS WRONG:

- ✗ **A)** This ignores the flat fee entirely.
- ✗ **B)** This swaps the rate and the flat fee — uses 3.50 as the rate per mile.
- ✓ **C)** CORRECT: flat fee $3.50 plus $1.75 per mile.
- ✗ **D)** This adds 1 dollar per mile instead of $1.75.

TRAP ALERT ■	Confusing which number is the rate vs. the fixed fee — students often swap them.
KEY INSIGHT ■	In y = mx + b word problems, identify the rate (m) and the starting value (b) separately.
FAST METHOD ■	Ask: what changes per unit? That's m. What's paid no matter what? That's b.

Q2. [EASY] **✓ Correct Answer: C) B = 0.10t + 25**

A cell phone plan costs $25 per month plus $0.10 per text message sent. If Priya sends t text messages in a month, which equation gives her total monthly bill B...

STEP-BY-STEP SOLUTION:

Monthly fee is $25 (fixed, doesn't depend on texts). Each text costs $0.10. Total: B = 0.10t + 25.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Ignores the $25 monthly fee.
- ✗ **B)** Treats $25 as the per-text rate, which is backwards.
- ✓ **C)** CORRECT: $0.10 per text plus $25 monthly base.
- ✗ **D)** Adds only 1 dollar per text.

TRAP ALERT ■	Students flip the per-unit rate and the fixed fee.
KEY INSIGHT ■	Fixed cost → b. Per-unit cost → m.
FAST METHOD ■	Check by substituting t = 0: should give B = 25 (the base fee alone).

Q3. [EASY] ✓ **Correct Answer: C) A = 6w + 18**

Marcus is saving money for a video game. He already has $18 saved and plans to save $6 each week. Which of the following equations models the total amount A, in...

STEP-BY-STEP SOLUTION:

Marcus already has $18 (initial value = 18). He saves $6 each week (rate = 6). After w weeks: A = 6w + 18.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Ignores the $18 already saved.
- ✗ **B)** Makes 18 the rate per week and 6 the starting amount — backwards.
- ✓ **C)** CORRECT: $6/week plus initial $18.
- ✗ **D)** Adds only $1 per week instead of $6.

TRAP ALERT ■	Forgetting the initial amount already saved.
KEY INSIGHT ■	'Already has' or 'starts with' = y-intercept (b).
FAST METHOD ■	Plug in w = 0 to verify: A = 18 ✓

Q4. [EASY] ✓ **Correct Answer: C) h = 12 – 0.5t**

A candle is 12 inches tall when lit. It burns at a rate of 0.5 inches per hour. Which equation gives the height h, in inches, of the candle after t hours of bur...

STEP-BY-STEP SOLUTION:

Initial height = 12 inches. The candle loses 0.5 inches per hour (decreasing). h = 12 – 0.5t.

WHY EACH OPTION IS WRONG:

- ✗ **A)** This adds height — candles don't grow!
- ✗ **B)** This adds height instead of subtracting.
- ✓ **C)** CORRECT: starts at 12 and decreases by 0.5 per hour.
- ✗ **D)** This has a negative starting height, which makes no sense.

TRAP ALERT ■	Using + instead of – for decreasing quantities.
KEY INSIGHT ■	Burning/draining/spending → subtract from the initial value.
FAST METHOD ■	At t = 2: h = 12 – 1 = 11 inches. Does that seem right? Yes.

Q5. [EASY] ✓ **Correct Answer: C) C = 40d + 15**

A car rental company charges $40 per day plus a one-time insurance fee of $15. Which equation represents the total cost C, in dollars, for renting a car for d d...

STEP-BY-STEP SOLUTION:

Daily rate = $40 per day (variable). Insurance = $15 (fixed, one-time). C = 40d + 15.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Ignores the $15 insurance fee.
- ✗ **B)** Makes 15 the per-day charge and 40 the fixed fee — reversed.
- ✓ **C)** CORRECT: $40 per day plus $15 flat fee.
- ✗ **D)** Adds both charges incorrectly into a single rate.

TRAP ALERT ■	Treating the one-time fee as the per-day rate.
KEY INSIGHT ■	One-time / flat / setup fee → b (y-intercept). Daily/per-unit → m (slope).
FAST METHOD ■	At d = 1: C = 40 + 15 = $55. Reasonable for one day with insurance.

Q6. [EASY] ✓ **Correct Answer: D) V = 8000 – 200h**

A swimming pool holds 8,000 gallons of water. Water drains from the pool at a rate of 200 gallons per hour. Which equation represents the volume V, in gallons, ...

STEP-BY-STEP SOLUTION:

Tank starts with 8,000 gallons. Drains 200 gallons per hour. V = 8000 – 200h.

WHY EACH OPTION IS WRONG:

- ✗ **A)** This represents only the drain amount, not the volume remaining.
- ✗ **B)** This multiplies 8,000 by hours — wrong structure.
- ✗ **C)** This adds water instead of draining it.
- ✓ **D)** CORRECT: starts at 8,000 and decreases by 200 each hour.

TRAP ALERT ■	Using + when the quantity is decreasing.
KEY INSIGHT ■	'Drains', 'empties', 'decreases' → subtract.
FAST METHOD ■	At h = 5: V = 8000 – 1000 = 7000 gallons. Makes sense.

Q7. [EASY] ✓ **Correct Answer: D) E = 12.50h + 30**

Kenji earns $12.50 per hour working at a bakery. Last week he also received a $30 tip. Which equation gives his total weekly earnings E, in dollars, if he worke...

STEP-BY-STEP SOLUTION:

Hourly wage = $12.50 × h. Plus one-time tip of $30. E = 12.50h + 30.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Ignores the $30 tip.
- ✗ **B)** Makes the tip the per-hour rate — backwards.
- ✗ **C)** Subtracts the tip instead of adding it.
- ✓ **D)** CORRECT: hourly earnings plus $30 tip.

TRAP ALERT ■	Subtracting the tip (treating it as a cost) instead of adding it.
KEY INSIGHT ■	Tips/bonuses/starting savings → add to total; they increase earnings.
FAST METHOD ■	At h = 4: E = 50 + 30 = $80. Reasonable.

Q8. [EASY] ✓ **Correct Answer: C) T = 2.75n + 10**

A store sells notebooks for $2.75 each. A customer also pays a one-time membership fee of $10. Which equation gives the total amount T, in dollars, a member pay...

STEP-BY-STEP SOLUTION:

$2.75 per notebook × n notebooks + $10 membership fee. T = 2.75n + 10.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Ignores the membership fee.
- ✗ **B)** Makes the membership fee the per-notebook cost.
- ✓ **C)** CORRECT: $2.75 per notebook plus $10 fixed membership.
- ✗ **D)** Adds both together into a single per-notebook price.

TRAP ALERT ■	Combining fixed and variable costs into one rate.
KEY INSIGHT ■	Always separate fixed (b) from per-unit (m) costs.
FAST METHOD ■	At n = 4: T = 11 + 10 = $21. Sensible.

Q9. [MEDIUM] ✓ **Correct Answer: A) V = 500 + 18t**

A water tank initially contains 500 liters. Water flows in at 30 liters per minute and flows out at 12 liters per minute. Which equation represents the volume V...

STEP-BY-STEP SOLUTION:

Net rate = inflow – outflow = 30 – 12 = 18 liters per minute (increasing). V = 500 + 18t.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: net gain of 18 L/min from the starting 500 L.
- ✗ **B)** This subtracts 18 — treats net flow as a loss.
- ✗ **C)** Adds the inflow only (30), ignoring the outflow.
- ✗ **D)** Subtracts total flow, ignoring inflow.

TRAP ALERT ■ Forgetting to subtract outflow from inflow to get the NET rate.

KEY INSIGHT ■ Net rate = inflow – outflow. If positive, tank is filling.

FAST METHOD ■ At t = 10: V = 500 + 180 = 680 L. Makes sense since water is entering faster than leaving.

Q10. [MEDIUM] ✓ **Correct Answer: C) C = 55h + 20**

A plumber charges $75 for the first hour and $55 for each additional hour. Which equation gives the total charge C, in dollars, for a job lasting h hours, where...

STEP-BY-STEP SOLUTION:

First hour costs $75. Each hour after that costs $55. For h total hours (h > 1): 1 hour at $75 + (h – 1) hours at $55 = 75 + 55(h – 1) = 75 + 55h – 55 = 55h + 20.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Adds $55 to $75 per hour — double-counts.
- ✗ **B)** Treats every hour as $55, missing the premium first hour.
- ✓ **C)** CORRECT: simplifies to 55h + 20.
- ✗ **D)** Combines both rates into $130 per hour incorrectly.

TRAP ALERT ■ Forgetting to simplify 75 + 55(h – 1): students leave it unsimplified and choose the wrong form.

KEY INSIGHT ■ Distribute first, then combine like terms when rewriting rate-change problems.

FAST METHOD ■ At h = 2: C = 75 + 55 = $130. Check: 55(2) + 20 = 130 ✓

Q11. **[MEDIUM]** **✓ Correct Answer: A) 200 + 8.50t = 624**

The cost of a school fundraiser dinner is $200 for venue rental plus $8.50 per ticket sold. The school collected $624 in ticket revenue. Which equation could be...

STEP-BY-STEP SOLUTION:

Total revenue = venue cost + ticket revenue. Venue = $200 (fixed). Ticket revenue = 8.50t. Set equal to $624: 200 + 8.50t = 624.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: fixed cost plus per-ticket revenue = total.
- ✗ **B)** Ignores the $200 venue cost.
- ✗ **C)** Makes $200 the per-ticket charge — wrong.
- ✗ **D)** Subtracts venue cost from ticket revenue — wrong structure.

TRAP ALERT ■ Omitting the fixed cost when writing the equation.

KEY INSIGHT ■ Total = fixed + variable. Don't leave out either piece.

FAST METHOD ■ Solve: 8.50t = 424 → t = about 50 tickets. Reasonable.

Q12. **[MEDIUM]** **✓ Correct Answer: A) s = 30t**

A rocket sled starts at rest and accelerates uniformly. After 3 seconds it is traveling at 90 feet per second, and after 7 seconds it is traveling at 210 feet p...

STEP-BY-STEP SOLUTION:

From t = 3 to t = 7: change in speed = 210 – 90 = 120; change in time = 4 sec. Rate = 120/4 = 30 ft/s per second. Using point (3, 90): 90 = 30(3) + b → b = 0. So s = 30t.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: slope = 30, starts from rest (b = 0).
- ✗ **B)** Uses wrong intercept.
- ✗ **C)** Gives negative speed at t = 0 — impossible for a sled starting from rest.
- ✗ **D)** Confuses the values of slope and intercept.

TRAP ALERT ■ Finding the slope correctly but using the wrong point to find the y-intercept.

KEY INSIGHT ■ Always verify by substituting both data points into your equation.

FAST METHOD ■ Check: s(3) = 90 ✓, s(7) = 210 ✓

Q13. [MEDIUM] ✓ **Correct Answer: A) F = 1.8C + 32**

A temperature reading in Celsius C and the corresponding Fahrenheit reading F satisfy a linear relationship. When C = 0, F = 32, and when C = 100, F = 212. Whic...

STEP-BY-STEP SOLUTION:

Two points: (0, 32) and (100, 212). Slope = (212 – 32)/(100 – 0) = 180/100 = 1.8. y-intercept = 32. F = 1.8C + 32.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: standard Celsius-to-Fahrenheit formula.
✗ **B)** Subtracts 32 instead of adding — wrong intercept.
✗ **C)** Reverses slope and intercept values.
✗ **D)** Wrong slope (0.8 instead of 1.8).

TRAP ALERT ■	Calculating slope as 100/180 instead of 180/100 (inverting rise and run).
KEY INSIGHT ■	Rise over run: change in F over change in C.
FAST METHOD ■	Verify: F(100) = 1.8(100) + 32 = 212 ✓

Q14. [MEDIUM] ✓ **Correct Answer: B) p = 22d**

Two friends each start reading a book on the same day. Anika starts on page 1 and reads 22 pages per day. Ben starts on page 50 (he had read ahead) and reads 15...

STEP-BY-STEP SOLUTION:

The question asks specifically for Anika's pages after d days. She reads 22 pages per day. After d days she has read 22d pages total.

WHY EACH OPTION IS WRONG:

✗ **A)** Adds 1 (her starting page position) which is a page number, not pages read.
✓ **B)** CORRECT: pages read by Anika = 22d.
✗ **C)** This is Ben's model.
✗ **D)** Subtracts a page that shouldn't be subtracted.

TRAP ALERT ■	Confusing 'page number reached' with 'pages read.' The question asks for total pages read.
KEY INSIGHT ■	Re-read the question: 'pages read' vs. 'page reached' are different quantities.
FAST METHOD ■	At d = 3: Anika reads 66 pages. Ben reads 15(3) = 45 pages from scratch + 50 ahead = 95 pages reached.

Q15. **[MEDIUM]**

✓ Correct Answer: C) The additional monthly cost per square foot of office space

A company rents office space. The monthly rent r, in dollars, is modeled by r = 1,200 + 4.50f, where f is the number of square feet rented. What is the best int...

STEP-BY-STEP SOLUTION:

In r = 1,200 + 4.50f, the slope 4.50 is the rate of change: for each additional square foot, the rent increases by $4.50.

WHY EACH OPTION IS WRONG:

- ✗ **A)** That would be the y-intercept ($1,200), not 4.50.
- ✗ **B)** 4.50 is the rate per square foot, not the total cost for just 1 sq ft (that would be $1,200 + $4.50 = $1,204.50).
- ✓ **C)** CORRECT: slope = rate of change = additional cost per square foot.
- ✗ **D)** There's no maximum implied by the slope value.

TRAP ALERT ■	Interpreting the slope as a fixed cost or total cost rather than a rate of change.
KEY INSIGHT ■	Slope = 'per unit' meaning. Always translate m as 'for each additional [x-unit], y changes by m'.
FAST METHOD ■	If f increases by 1, r increases by exactly $4.50. That's what 'additional cost per square foot' means.

Q16. **[MEDIUM]**

✓ Correct Answer: A) The athlete's heart rate at the start of the workout

An athlete's heart rate H, in beats per minute, during a workout session is modeled by H = 72 + 5.4m, where m is the number of minutes into the workout. What do...

STEP-BY-STEP SOLUTION:

In H = 72 + 5.4m, when m = 0 (start of workout), H = 72. The y-intercept is the initial value.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: at m = 0, H = 72 — resting/starting heart rate.
- ✗ **B)** That's the slope (5.4 bpm increase per minute).
- ✗ **C)** Maximum is not given by the y-intercept.
- ✗ **D)** Heart rate = 0 is not physiologically meaningful here.

TRAP ALERT ■	Confusing the y-intercept (initial value) with the slope (rate of change).
KEY INSIGHT ■	y-intercept = value when x = 0. Slope = change per unit of x.
FAST METHOD ■	H(0) = 72. H(1) = 77.4. The jump of 5.4 per minute is the rate (slope).

Q17. **[MEDIUM]** ✓ **Correct Answer: A) $C = 10g + 160$**

A catering service charges by the number of guests. For a party with 80 guests, the charge is $960. For a party with 120 guests, the charge is $1,360. Assuming ...

STEP-BY-STEP SOLUTION:

Two points: (80, 960) and (120, 1360). Slope = (1360 – 960)/(120 – 80) = 400/40 = 10. Using (80, 960): 960 = 10(80) + b → b = 160. C = 10g + 160.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: slope 10, intercept 160.
- ✗ **B)** Slope of 8 gives 8(80) = 640 + b = 960 → b = 320, but check: 8(120) + 320 = 1280 ≠ 1360. Wrong.
- ✗ **C)** Slope of 12: 12(80) = 960, so b = 0, but 12(120) = 1440 ≠ 1360. Wrong.
- ✗ **D)** Intercept error.

TRAP ALERT ■	Computing slope correctly but then making an arithmetic error finding the y-intercept.
KEY INSIGHT ■	Once you have the slope, substitute ONE data point carefully: C – mg = b.
FAST METHOD ■	Verify both points: 10(80)+160=960 ✓, 10(120)+160=1360 ✓

Q18. **[MEDIUM]** ✓ **Correct Answer: A) $d = 9t$**

A diver descends from the surface at a constant rate. After 2 minutes the diver is at a depth of 18 meters, and after 5 minutes the diver is at 45 meters. Which...

STEP-BY-STEP SOLUTION:

Points: (2, 18) and (5, 45). Slope = (45 – 18)/(5 – 2) = 27/3 = 9. y-intercept: 18 = 9(2) + b → b = 0. d = 9t.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: slope = 9, starts at surface (d = 0 when t = 0).
- ✗ **B)** Negative intercept would mean diver started above the surface.
- ✗ **C)** Slope of 18 gives too steep a rate; check: d(2) = 36 ≠ 18.
- ✗ **D)** Adds 2 to time — wrong manipulation.

TRAP ALERT ■	Getting slope right but adding a nonzero intercept by using coordinates incorrectly.
KEY INSIGHT ■	Check that d = 0 when t = 0 (diver starts at surface). If so, b = 0.
FAST METHOD ■	d(5) = 9(5) = 45 ✓

Q19. **[MEDIUM]** ✓ **Correct Answer: A) About 90 loaves**

The equation P = 4.20n – 380 models the monthly profit P, in dollars, for a bakery that sells n loaves of bread per month. According to this model, how many loa...

STEP-BY-STEP SOLUTION:

Set P = 0: 4.20n – 380 = 0 → 4.20n = 380 → n = 380/4.20 ≈ 90.48. So about 90 loaves (round up to 91 for actual break-even, but 'about 90' is closest).

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: 380/4.20 ≈ 90.5 ≈ 90 loaves.
- ✗ **B)** 95 would be 4.20(95) – 380 = 399 – 380 = 19 > 0. Closer than needed.
- ✗ **C)** 76 is too few: 4.20(76) = 319.2 – 380 = –60.8 (loss).
- ✗ **D)** 80 loaves: 4.20(80) – 380 = 336 – 380 = –44 (still a loss).

TRAP ALERT ■	Dividing 380 by 4.20 incorrectly (inverting the fraction).
KEY INSIGHT ■	Break-even: set profit = 0 and solve for n. Divide the fixed cost by the per-unit profit margin.
FAST METHOD ■	380 ÷ 4.20 = 90.47... → at least 91 loaves to actually profit.

Q20. **[MEDIUM]** ✓ **Correct Answer: A) 120 + 35j < 45j**

A landscaping crew earns a set fee of $120 plus $35 per hour for additional labor. If a second crew earns only $45 per hour with no set fee, and both crews work...

STEP-BY-STEP SOLUTION:

First crew total: 120 + 35j. Second crew total: 45j. For first crew to charge less: 120 + 35j < 45j.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: first crew cost < second crew cost.
- ✗ **B)** This means first crew costs MORE — opposite of what's asked.
- ✗ **C)** Wrong structure — subtracts the base fee from the wrong side.
- ✗ **D)** Flips the fixed fee sign.

TRAP ALERT ■	Writing > instead of < (mixing up 'charges less' vs. 'charges more').
KEY INSIGHT ■	'Charges less than' = <. 'Charges more than' = >.
FAST METHOD ■	Solving: 120 < 10j → j > 12. So after 12 hours, crew A is cheaper.

Q21. **[MEDIUM]** **✓ Correct Answer: C) The number of hours until the tank is empty**

The function f(x) = –3.2x + 96 models the amount of fuel f (in gallons) remaining in a truck's tank after x hours of driving. What is the most reasonable interp...

STEP-BY-STEP SOLUTION:

x-intercept: set f(x) = 0. 0 = –3.2x + 96 → x = 96/3.2 = 30. After 30 hours, the tank is empty. The x-intercept represents 'when the fuel runs out.'

WHY EACH OPTION IS WRONG:

- ✗ **A)** Initial fuel = f(0) = 96 gallons — that's the y-intercept, not x-intercept.
- ✗ **B)** Rate of consumption is the slope magnitude (3.2 gal/hr).
- ✓ **C)** CORRECT: x-intercept = time when f = 0 (tank empty).
- ✗ **D)** f(1) = 96 – 3.2 = 92.8 gallons (still remaining, not zero).

TRAP ALERT ■	Confusing x-intercept with y-intercept interpretations.
KEY INSIGHT ■	x-intercept: set y = 0 → find when the OUTPUT is zero. y-intercept: set x = 0 → find the INITIAL value.
FAST METHOD ■	96 ÷ 3.2 = 30 hours. The truck runs dry after 30 hours of driving.

Q22. **[MEDIUM]** **✓ Correct Answer: C) m = 5**

A gym membership costs $50 to join and $30 per month. A competing gym has no joining fee but costs $40 per month. After how many months m will the total cost of...

STEP-BY-STEP SOLUTION:

Gym 1: 50 + 30m. Gym 2: 40m. Set equal: 50 + 30m = 40m → 50 = 10m → m = 5.

WHY EACH OPTION IS WRONG:

- ✗ **A)** m = 3: Gym 1 = $140, Gym 2 = $120. Not equal.
- ✗ **B)** m = 4: Gym 1 = $170, Gym 2 = $160. Not equal.
- ✓ **C)** CORRECT: m = 5 gives both = $200.
- ✗ **D)** m = 6: Gym 1 = $230, Gym 2 = $240. Already past equal point.

TRAP ALERT ■	Forgetting to include the joining fee when setting up the equation.
KEY INSIGHT ■	Set total costs equal. Solve for the variable. Then verify by substituting back.
FAST METHOD ■	50 + 30(5) = 200. 40(5) = 200 ✓

Q23. [MEDIUM] ✓ **Correct Answer: A) O = 200 – 6t**

A new apartment building has 200 units. The property manager expects that d units will be vacant each month due to tenant turnover. After 6 months, only 164 uni...

STEP-BY-STEP SOLUTION:

After 6 months: 200 – O = 200 – 164 = 36 units became vacant. Rate = 36/6 = 6 units per month. O = 200 – 6t.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: starts at 200, loses 6 per month.
✗ **B)** Rate of 36 per month is far too high (entire building gone in ~5.5 months).
✗ **C)** This adds units — occupied units would grow, which contradicts turnover.
✗ **D)** Also adds units — wrong direction.

TRAP ALERT ■	Using the total change over all months as the monthly rate (36 instead of 36/6 = 6).
KEY INSIGHT ■	Rate = total change ÷ number of time periods.
FAST METHOD ■	At t = 6: O = 200 – 36 = 164 ✓

Q24. [HARD] ✓ **Correct Answer: B) P = 2.70w – 420**

A factory produces widgets at a cost of $3.80 per widget plus fixed daily costs of $420. The factory sells each widget for $6.50. Which of the following equatio...

STEP-BY-STEP SOLUTION:

Revenue per widget = $6.50. Cost per widget = $3.80. Profit per widget = 6.50 – 3.80 = $2.70. Fixed daily cost = $420. P = 2.70w – 420.

WHY EACH OPTION IS WRONG:

✗ **A)** Subtracts only the fixed cost without netting the variable cost per unit.
✓ **B)** CORRECT: profit margin × units – fixed cost.
✗ **C)** Uses only the variable cost and adds the fixed cost.
✗ **D)** Gets the profit margin right but adds the fixed cost instead of subtracting.

TRAP ALERT ■	Forgetting that profit = revenue – ALL costs (both variable and fixed).
KEY INSIGHT ■	Profit = (sell price – cost price) × units – fixed costs.
FAST METHOD ■	At w = 200: P = 2.70(200) – 420 = 540 – 420 = $120 ✓

Q25. [HARD] ✓ **Correct Answer: D) t = 10**

The linear model V = 24,000 – 1,800t represents the value V, in dollars, of a car t years after purchase. A second car is purchased at the same time for $18,000...

STEP-BY-STEP SOLUTION:

Car 1: V■ = 24,000 – 1,800t. Car 2: V■ = 18,000 – 1,200t. Set equal: 24,000 – 1,800t = 18,000 – 1,200t → 6,000 = 600t → t = 10. Verify: V■(10) = 24,000 – 18,000 = 6,000. V■(10) = 18,000 – 12,000 = 6,000. Both equal $6,000 ✓.

WHY EACH OPTION IS WRONG:

- ✗ **A)** t = 4: Car 1 = $16,800; Car 2 = $13,200. Not equal.
- ✗ **B)** t = 5: Car 1 = $15,000; Car 2 = $12,000. Not equal.
- ✗ **C)** t = 6: Car 1 = $13,200; Car 2 = $10,800. Not equal.
- ✓ **D)** CORRECT: t = 10. Both = $6,000.

TRAP ALERT ■	Setting up the equation correctly but making arithmetic errors subtracting terms with different coefficients.
KEY INSIGHT ■	After setting equations equal, isolate t systematically: move t-terms to one side, constants to the other.
FAST METHOD ■	24,000 – 18,000 = 1,800t – 1,200t → 6,000 = 600t → t = 10 ✓

Q26. [HARD] ✓ **Correct Answer: A) m = –2.4t + 50**

A scientist measures the mass of a radioactive sample every hour. The data suggest a linear relationship for the first 10 hours: at t = 1 the mass is 47.6 grams...

STEP-BY-STEP SOLUTION:

Points: (1, 47.6) and (4, 40.4). Slope = (40.4 – 47.6)/(4 – 1) = –7.2/3 = –2.4. Using point (1, 47.6): 47.6 = –2.4(1) + b → b = 50. m = –2.4t + 50.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: slope –2.4, intercept 50.
- ✗ **B)** Wrong intercept (40 instead of 50).
- ✗ **C)** Positive slope — mass would increase (wrong direction).
- ✗ **D)** Uses total change (–7.2) instead of per-hour rate.

TRAP ALERT ■	Using the total change between the two points (–7.2) as the slope instead of dividing by the time interval (3).
KEY INSIGHT ■	Slope = Δm / Δt, NOT just Δm.
FAST METHOD ■	Verify: m(1) = –2.4 + 50 = 47.6 ✓, m(4) = –9.6 + 50 = 40.4 ✓

Q27. [HARD] ✓ **Correct Answer: D) u > 1,800**

Company A charges $0.15 per unit plus a monthly fee of $180. Company B charges $0.25 per unit with no monthly fee. For what number of units u is Company A's mon...

STEP-BY-STEP SOLUTION:

Company A: 0.15u + 180. Company B: 0.25u. For A < B: 0.15u + 180 < 0.25u → 180 < 0.10u → u > 1,800.

WHY EACH OPTION IS WRONG:

- ✗ **A)** u < 1,200 is incorrect — threshold is 1,800, not 1,200, and direction is >.
- ✗ **B)** u < 1,800 reverses the inequality — Company B would be cheaper at u < 1,800.
- ✗ **C)** u > 1,200 uses the wrong threshold.
- ✓ **D)** CORRECT: at more than 1,800 units, Company A's total cost is less than Company B's.

TRAP ALERT ■	Flipping the inequality sign when dividing or subtracting variable terms.
KEY INSIGHT ■	Move ALL u-terms to one side before solving. Check with a test value.
FAST METHOD ■	At u = 2,000: A = 300 + 180 = $480; B = 500. A < B ✓ So u > 1,800 makes A cheaper.

Q28. [HARD] ✓ **Correct Answer: B) More than 6,000**

The equation C = 0.08s + 1,200 models the monthly operating cost C, in dollars, of running a delivery service, where s is the number of packages delivered that ...

STEP-BY-STEP SOLUTION:

C_service = 0.08s + 1,200. C_rival = 1,680. For rival to be cheaper: 1,680 < 0.08s + 1,200 → 480 < 0.08s → s > 6,000.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 5,000: 0.08(5000) + 1200 = 400 + 1200 = $1,600 < $1,680. Service is still cheaper.
- ✓ **B)** CORRECT: 6,001 packages makes service cost $1,680.08 > rival's $1,680.
- ✗ **C)** 7,000: Service = $1,760. Rival cheaper, but threshold is 6,000.
- ✗ **D)** Less than 6,000 makes the service cheaper, not rival.

TRAP ALERT ■	Comparing the wrong direction: students sometimes find when the SERVICE is cheaper instead of when the RIVAL is cheaper.
KEY INSIGHT ■	Re-read: 'makes the rival service cheaper' means rival cost < service cost.
FAST METHOD ■	At s = 6,001: service = $1,680.08, rival = $1,680. Rival is $0.08 cheaper ✓

Q29. [HARD] ✓ **Correct Answer: B) E = 3,200 + 0.04(s – 10,000)**

An employee receives a base salary of $3,200 per month plus a commission of 4% on all sales above $10,000. Which of the following equations gives the employee's...

STEP-BY-STEP SOLUTION:

Base salary = $3,200. Commission = 4% of sales ABOVE $10,000 = 0.04(s – 10,000). Total: E = 3,200 + 0.04(s – 10,000).

WHY EACH OPTION IS WRONG:

- ✗ **A)** Applies 4% to ALL sales, not just the amount above $10,000.
- ✓ **B)** CORRECT: commission only on (s – 10,000).
- ✗ **C)** Adds $10,000 instead of subtracting — inflates commission incorrectly.
- ✗ **D)** Subtracts the base salary — wrong sign.

TRAP ALERT ■	Applying the commission to total sales instead of to sales ABOVE the threshold.
KEY INSIGHT ■	'Above $10,000' means subtract $10,000 from s before multiplying by the commission rate.
FAST METHOD ■	If s = $15,000: commission = 0.04(5,000) = $200. E = $3,400 ✓

Q30. [HARD] ✓ **Correct Answer: A) d = 310 – 50t**

A train travels at a constant speed. At 9:00 AM it is 310 miles from its destination, and at 11:30 AM it is 185 miles from its destination. Which linear equatio...

STEP-BY-STEP SOLUTION:

At 9 AM (t = 0): d = 310. At 11:30 AM (t = 2.5): d = 185. Slope = (185 – 310)/2.5 = –125/2.5 = –50. d = 310 – 50t.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: starts at 310 miles, closes at 50 mph.
- ✗ **B)** Adds distance instead of subtracting.
- ✗ **C)** Uses 185 as the start — wrong.
- ✗ **D)** Negative starting distance is impossible.

TRAP ALERT ■	Converting 11:30 AM to t = 2.5 hours incorrectly (e.g., using t = 2 hours 30 minutes = 230 instead of 2.5).
KEY INSIGHT ■	Always convert time to decimal hours: 30 minutes = 0.5 hours.
FAST METHOD ■	At t = 2.5: d = 310 – 125 = 185 ✓

Q31. [HARD] ✓ **Correct Answer: B) At least 10 meals per day**

The monthly profit P of a small restaurant is modeled by P = 18.50n – 3,700, where n is the number of meals served. If the restaurant is open 30 days per month,...

STEP-BY-STEP SOLUTION:

Set $P \geq 1{,}850$: $18.50n - 3{,}700 \geq 1{,}850 \rightarrow 18.50n \geq 5{,}550 \rightarrow n \geq 300$ meals per month. Per day: 300/30 = 10 meals per day.

WHY EACH OPTION IS WRONG:

✗ **A)** 15 meals/day × 30 = 450 meals → P = 18.5(450) – 3700 = $4,625. More than needed, not minimum.
✓ **B)** CORRECT: 10 meals/day × 30 = 300 meals → P = 18.5(300) – 3700 = $1,850. Exactly meets target.
✗ **C)** 22/day → 660 meals: far above what's needed.
✗ **D)** 42/day overshoots significantly.

TRAP ALERT ■	Forgetting to divide the monthly meal count by the number of days open.
KEY INSIGHT ■	Monthly requirement ÷ days open = daily requirement.
FAST METHOD ■	300 meals/month ÷ 30 days = 10 meals/day ✓

Q32. [HARD] ✓ **Correct Answer: A) y-intercept = 15**

A linear function passes through (2, 11) and (6, 3). What is the y-intercept of this function?

STEP-BY-STEP SOLUTION:

Points: (2, 11) and (6, 3). Slope = $(3 - 11)/(6 - 2) = -8/4 = -2$. Using (2, 11): $11 = -2(2) + b \rightarrow 11 = -4 + b \rightarrow b = 15$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: b = 15.
✗ **B)** b = 13 would come from an arithmetic error (+2 instead of –2 for slope).
✗ **C)** b = 17 comes from subtracting instead of adding 4.
✗ **D)** b = 9 is incorrect.

TRAP ALERT ■	Sign errors when computing b: students sometimes write $11 = -2(2) + b \rightarrow 11 = 4 + b \rightarrow b = 7$ (wrong sign on -2×2).
KEY INSIGHT ■	$-2 \times 2 = -4$, not +4. Carry the negative.
FAST METHOD ■	Verify: $f(6) = -2(6) + 15 = -12 + 15 = 3$ ✓

Q33. [HARD] ✓ **Correct Answer: A) T = 160**

On a highway, speed limit signs are posted every k miles. The number of signs N within a 240-mile stretch is modeled by N = 240/k. If the equation were rewritte...

STEP-BY-STEP SOLUTION:

With $k = 3$: $N = 240/3 = 80$ signs. Each sign takes 2 seconds. $T = 80 \times 2 = 160$ seconds.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: 80 signs × 2 sec = 160 seconds.
- ✗ **B)** 480 = 240 × 2. Treats entire 240-mile stretch as one unit.
- ✗ **C)** 80 would be T if each sign took 1 second.
- ✗ **D)** 240 ignores the 2-second reading time.

TRAP ALERT ■ Forgetting to multiply by 2 seconds per sign after finding the number of signs.

KEY INSIGHT ■ Two-step: (1) find N, then (2) multiply by time per sign.

FAST METHOD ■ $N = 240/3 = 80$; $T = 80 \times 2 = 160$ ✓

Q34. [HARD] ✓ **Correct Answer: B) 210 units of Product B**

A manufacturer produces two products. Product A yields $5 profit per unit and Product B yields $8 profit per unit. The total weekly profit is $2,480. If the man...

STEP-BY-STEP SOLUTION:

$5(160) + 8B = 2{,}480 \rightarrow 800 + 8B = 2{,}480 \rightarrow 8B = 1{,}680 \rightarrow B = 210.$

WHY EACH OPTION IS WRONG:

- ✗ **A)** 200 units: $5(160) + 8(200) = 800 + 1600 = 2400 \neq 2480$.
- ✓ **B)** CORRECT: $5(160) + 8(210) = 800 + 1680 = 2480$ ✓.
- ✗ **C)** 180 units: $800 + 1440 = 2240 \neq 2480$.
- ✗ **D)** 220 units: $800 + 1760 = 2560 \neq 2480$.

TRAP ALERT ■ Dividing 1,680 by 5 (Product A's profit) instead of 8 (Product B's profit).

KEY INSIGHT ■ Set up: $5(160) + 8B = 2{,}480$. Solve for B step by step.

FAST METHOD ■ $8B = 2480 - 800 = 1680 \rightarrow B = 1680/8 = 210$ ✓

Q35. [HARD] ✓ **Correct Answer: A) V = 4,500 – 60d**

Water evaporates from a reservoir at a linear rate. On Day 0, the reservoir contains 4,500 acre-feet. On Day 12, it contains 3,780 acre-feet. Which equation giv...

STEP-BY-STEP SOLUTION:

At d = 0: V = 4,500. At d = 12: V = 3,780. Change = 4,500 – 3,780 = 720 over 12 days. Rate = 720/12 = 60 acre-feet per day. V = 4,500 – 60d.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: starts at 4,500, loses 60 per day.
- ✗ **B)** Adds water — wrong direction.
- ✗ **C)** Uses Day 12 value as starting point — wrong.
- ✗ **D)** Uses total loss (720) as daily rate — off by factor of 12.

TRAP ALERT ■	Using total evaporation (720) instead of daily evaporation rate (720 ÷ 12 = 60).
KEY INSIGHT ■	Rate = total change ÷ number of days.
FAST METHOD ■	V(12) = 4500 – 60(12) = 4500 – 720 = 3780 ✓

Q36. [MEDIUM] ✓ **Correct Answer: A) a + s = 220 and 14a + 8s = 2,480**

A local theater sells adult tickets for $14 and student tickets for $8. At one performance, 220 tickets were sold for a total revenue of $2,480. Which system of...

STEP-BY-STEP SOLUTION:

Total tickets: a + s = 220. Total revenue: 14a + 8s = 2,480.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: both equations are properly set up.
- ✗ **B)** Reverses the roles of 220 and 2,480.
- ✗ **C)** Uses a minus s — no reason to subtract ticket counts.
- ✗ **D)** Swaps the prices of adult and student tickets.

TRAP ALERT ■	Writing the revenue equation with swapped prices (8 for adults, 14 for students).
KEY INSIGHT ■	Match each variable to its price carefully. Adult = a = $14; student = s = $8.
FAST METHOD ■	This is a classic 2-variable linear system. Set up carefully before solving.

Q37. [MEDIUM] ✓ **Correct Answer: B) d = 4(t + 1/3)**

Natalie walks from home to school at 4 miles per hour and her brother drives the same route at 30 miles per hour. Natalie leaves 20 minutes before her brother. ...

STEP-BY-STEP SOLUTION:

Natalie leaves 20 minutes = 1/3 hour before her brother starts. So by the time the brother leaves, Natalie has been walking for t + (1/3) hours. d = 4(t + 1/3), where t = 0 when the brother leaves.

WHY EACH OPTION IS WRONG:

- ✗ **A)** d = 4t ignores her head start.
- ✓ **B)** CORRECT: accounts for her 20-minute (1/3 hour) head start.
- ✗ **C)** That's the brother's speed — wrong person.
- ✗ **D)** Subtracts 20 minutes rather than converting to hours and adding.

TRAP ALERT ■	Subtracting 20 instead of adding 1/3 hour (incorrect direction and unit conversion).
KEY INSIGHT ■	20 minutes = 20/60 = 1/3 hour. Natalie has been walking longer, so add the head start to t.
FAST METHOD ■	At t = 0 (when brother leaves): d = 4(1/3) = 4/3 miles. Natalie has a head start ✓

Q38. [MEDIUM] ✓ **Correct Answer: B) S = 300n + 6,900**

A fitness app tracks a user's daily step count. On Monday the user took 7,200 steps, and each subsequent day the user increased steps by 300. Which linear equat...

STEP-BY-STEP SOLUTION:

On Day 1 (n = 1): S = 7,200. This is the first term. The pattern increases by 300 each day. S = 7,200 + 300(n – 1) = 7,200 + 300n – 300 = 300n + 6,900.

WHY EACH OPTION IS WRONG:

- ✗ **A)** S = 300n + 7,200 gives S(1) = 7,500 ≠ 7,200. Off by one step.
- ✓ **B)** CORRECT: S(1) = 300 + 6,900 = 7,200 ✓
- ✗ **C)** 7,200n is not the right structure.
- ✗ **D)** Subtracts — steps would decrease.

TRAP ALERT ■	Using n = 0 for the first day instead of n = 1, causing an off-by-one error in the intercept.
KEY INSIGHT ■	When n = 1 is Day 1, the formula is a + (n – 1)d, which simplifies by distributing.
FAST METHOD ■	S(1) = 7200, S(2) = 7500, S(3) = 7800... fits B.

Q39. **[MEDIUM]** **✓ Correct Answer: A) $760**

A moving company charges $90 per hour for a crew of 3 workers plus $0.50 per pound for items above 2,000 pounds. A move takes 4 hours and the items weigh 2,800 ...

STEP-BY-STEP SOLUTION:

Crew time: 4 hours × $90/hr = $360. Overweight: 2,800 – 2,000 = 800 lbs over limit. Surcharge: 800 × $0.50 = $400. Total: $360 + $400 = $760.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $360 + $400 = $760.
- ✗ **B)** 'Plus tax' is not in the problem.
- ✗ **C)** $400 is only the overweight charge — missing labor.
- ✗ **D)** $1,000 adds a nonexistent component.

TRAP ALERT ■	Forgetting one of the two charges (labor or overweight surcharge).
KEY INSIGHT ■	List ALL charges, compute each, then sum them.
FAST METHOD ■	Hourly labor: 4 × $90 = $360. Overweight: 800 × $0.50 = $400. Total: $760 ✓

Q40. **[MEDIUM]** **✓ Correct Answer: C) A fixed setup fee charged regardless of how many shirts are printed**

The cost C, in dollars, to print custom t-shirts at a shop is C = 8.25n + 45, where n is the number of shirts. Which statement best describes the 45 in this equ...

STEP-BY-STEP SOLUTION:

In C = 8.25n + 45, the 45 is the y-intercept (value when n = 0). This represents a constant, one-time charge regardless of quantity.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Per-shirt cost is 8.25 (the slope), not 45.
- ✗ **B)** The total cost for 45 shirts would be 8.25(45) + 45 = $416.25, not $45.
- ✓ **C)** CORRECT: 45 is the fixed setup/base fee.
- ✗ **D)** n can go above 45; it's not a maximum.

TRAP ALERT ■	Confusing the y-intercept dollar value with a quantity (number of shirts).
KEY INSIGHT ■	y-intercept = fixed cost (what you pay when n = 0). Slope = per-unit cost.
FAST METHOD ■	C(0) = $45 even with zero shirts. That's a setup fee ✓

Q41. [HARD] ✓ **Correct Answer: A) After 8 hours**

Two workers are painting a house. Worker A has already painted 120 square feet and paints at 40 square feet per hour. Worker B starts at the same time as Worker...

STEP-BY-STEP SOLUTION:

Worker A: 120 + 40t. Worker B: 55t. Set B > A: 55t > 120 + 40t → 15t > 120 → t > 8.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: after t > 8 hours, Worker B overtakes Worker A.
✗ **B)** After 9 is also true, but the crossover is at 8 hours.
✗ **C)** At t = 7: A = 400, B = 385. B is still behind.
✗ **D)** After 10 is later than necessary.

TRAP ALERT ■	Writing A > B instead of B > A (getting the inequality direction reversed).
KEY INSIGHT ■	'Worker B exceeds Worker A' means B's total > A's total.
FAST METHOD ■	At t = 8: A = 120 + 320 = 440; B = 55(8) = 440. Equal at t = 8, so B passes A after t = 8.

Q42. [HARD] ✓ **Correct Answer: C) The person's weight at the start of the diet**

The linear model W = 185 – 2.5t represents a person's weight W in pounds t weeks after starting a diet. What is the most reasonable interpretation of the y-inte...

STEP-BY-STEP SOLUTION:

In W = 185 – 2.5t, when t = 0 (start of diet), W = 185. The y-intercept is the initial weight.

WHY EACH OPTION IS WRONG:

✗ **A)** W after 2.5 weeks = 185 – 6.25 = 178.75 lb. Not the y-intercept.
✗ **B)** The rate of weight loss is the slope magnitude (2.5 lb/week).
✓ **C)** CORRECT: W(0) = 185 is the initial weight.
✗ **D)** Total weeks is not given by any single coefficient here.

TRAP ALERT ■	Interpreting the y-intercept as a rate or duration rather than an initial value.
KEY INSIGHT ■	y-intercept = starting value. Slope = rate of change (negative here means loss).
FAST METHOD ■	At t = 0: W = 185. First week: W = 182.5. Decreasing at 2.5 lb/week.

Q43. [HARD] ✓ **Correct Answer: A) F = 28 – (2/3)s**

An elevator starts at floor 28 and descends at a constant rate of 2 floors every 3 seconds. A second elevator starts at floor 4 and ascends at a constant rate o...

STEP-BY-STEP SOLUTION:

Descends 2 floors every 3 seconds → rate = 2/3 floor per second (negative since descending). Starts at floor 28. F = 28 – (2/3)s.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: starts at floor 28, descends at 2/3 floor/sec.
- ✗ **B)** This ascends — wrong direction for descent.
- ✗ **C)** Rate of 2 floors/sec is too fast (that's 2 floors per 1 second, not per 3).
- ✗ **D)** Negative start — impossible.

TRAP ALERT ■	Incorrectly computing the rate as 2 floors/sec instead of 2/3 floors/sec.
KEY INSIGHT ■	'2 floors every 3 seconds' = 2/3 floor per second. Don't drop the denominator.
FAST METHOD ■	At s = 3: F = 28 – 2 = 26 (descended 2 floors in 3 seconds) ✓

Q44. [HARD] ✓ **Correct Answer: C) 2026**

A school district's budget B, in millions of dollars, follows the model B = 42 + 1.8y, where y is the number of years after 2015. According to this model, in wh...

STEP-BY-STEP SOLUTION:

Set B > 60: 42 + 1.8y > 60 → 1.8y > 18 → y > 10. At y = 10: B = 42 + 18 = 60 (not exceeding, exactly equal). At y = 11: B = 42 + 19.8 = 61.8 > 60. Year = 2015 + 11 = 2026.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 2024 = y = 9: B = 42 + 16.2 = 58.2 < 60.
- ✗ **B)** 2025 = y = 10: B = 42 + 18 = 60. Equals 60 but does not exceed it.
- ✓ **C)** CORRECT: 2026 = y = 11: B = 42 + 19.8 = 61.8 > 60. First year budget exceeds $60M.
- ✗ **D)** 2027 = y = 12: B = 63.6 > 60 but not the FIRST year to exceed.

TRAP ALERT ■	Finding when B = 60 (not exceeding) instead of when B > 60 (strictly exceeding).
KEY INSIGHT ■	'First exceed' means the first year B > 60 (strictly greater than). Check the next year after equality.
FAST METHOD ■	y = 10 → B = exactly 60 (not exceeded). y = 11 → B = 61.8 > 60 ✓ Answer: 2026.

Q45. **[HARD]** **✓ Correct Answer: A) A = 5 – 0.4m**

A lab starts an experiment with 5 grams of a substance. The substance dissolves at a constant rate. After 8 minutes, 1.8 grams remain. Which linear equation mod...

STEP-BY-STEP SOLUTION:

At $m = 0$: $A = 5$ g. At $m = 8$: $A = 1.8$ g. Slope $= (1.8 - 5)/(8 - 0) = -3.2/8 = -0.4$. $A = 5 - 0.4m$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: starts at 5g, dissolves at 0.4 g/min.
- ✗ **B)** Positive slope — substance would increase. Impossible for dissolving.
- ✗ **C)** Uses 1.8 as starting point — wrong.
- ✗ **D)** Rate of 0.8 gives $A(8) = 5 - 6.4 = -1.4 < 0$. Impossible.

TRAP ALERT ■	Computing slope as total change (–3.2) instead of change per minute (–3.2 / 8 = –0.4).
KEY INSIGHT ■	Slope = $\Delta y / \Delta x$ = change in mass / change in time.
FAST METHOD ■	$A(8) = 5 - 0.4(8) = 5 - 3.2 = 1.8$ ✓

Q46. **[HARD]** **✓ Correct Answer: A) w > 20**

A delivery service offers two pricing plans. Plan X: $2.50 per pound with no base charge. Plan Y: $1.80 per pound plus a $14 base charge. For what weight w, in ...

STEP-BY-STEP SOLUTION:

Plan X: $2.50w$. Plan Y: $1.80w + 14$. For Y < X: $1.80w + 14 < 2.50w \rightarrow 14 < 0.70w \rightarrow w > 20$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: above 20 lbs, Plan Y is cheaper.
- ✗ **B)** $w < 20$ makes Plan X cheaper.
- ✗ **C)** $w > 14$ is not the correct threshold.
- ✗ **D)** $w > 10$ is too small; check: $1.80(10) + 14 = 32$ vs $2.50(10) = 25$. Plan Y costs more at 10 lbs.

TRAP ALERT ■	Solving $14 < 0.70w$ as $w > 14$ (dividing by 1 instead of 0.70).
KEY INSIGHT ■	Divide both sides by 0.70: $w > 14/0.70 = 20$.
FAST METHOD ■	At $w = 21$: X = $52.50, Y = $51.80. Y is cheaper ✓

Q47. **[HARD]**	**✓ Correct Answer: C) The distance from the peak at the start of the descent**

The function $f(t) = 3{,}600 - 45t$ models the distance in meters between a hiker and a mountain peak t minutes after the hiker begins the descent. What does the va...

STEP-BY-STEP SOLUTION:

$f(0) = 3{,}600 - 45(0) = 3{,}600$ meters. This is the initial distance from the peak — the y-intercept.

WHY EACH OPTION IS WRONG:

- ✗ **A)** The hiker's speed is 45 m/min (the slope magnitude), not 3,600.
- ✗ **B)** Time to reach base: 3,600/45 = 80 minutes. That's the x-intercept.
- ✓ **C)** CORRECT: y-intercept = initial distance = 3,600 m.
- ✗ **D)** $f(1) = 3{,}555$ — that's distance remaining after 1 minute, not what 3,600 means.

TRAP ALERT ■	Confusing the y-intercept (initial distance) with the slope (speed).
KEY INSIGHT ■	y-intercept = starting condition ($t = 0$). Slope = rate of change.
FAST METHOD ■	$f(0) = 3{,}600$. The hiker starts 3,600 m from the peak.

Q48. **[HARD]**	**✓ Correct Answer: C) 16 thousand phones**

A phone manufacturer's revenue R, in millions, is modeled by $R = 12n - 180$, where n is the number of phones (in thousands) produced and sold. What is the minimu...

STEP-BY-STEP SOLUTION:

Set $R > 0$: $12n - 180 > 0 \rightarrow 12n > 180 \rightarrow n > 15$. The smallest integer greater than 15 is 16. At $n = 16$: $R = 12(16) - 180 = 192 - 180 = 12 > 0$.

WHY EACH OPTION IS WRONG:

- ✗ **A)** $n = 14$: $R = 12(14) - 180 = 168 - 180 = -12 < 0$. Negative revenue.
- ✗ **B)** $n = 15$: $R = 12(15) - 180 = 180 - 180 = 0$. Break-even, but not positive.
- ✓ **C)** CORRECT: $n = 16 \rightarrow R = 12 > 0$. First value giving positive revenue.
- ✗ **D)** $n = 18 \rightarrow R = 36$. Positive but not the minimum.

TRAP ALERT ■	Choosing the break-even point ($n = 15$, $R = 0$) instead of the first value where R is strictly positive.
KEY INSIGHT ■	'Positive' means > 0 (strictly). Break-even (= 0) is NOT positive. Go one unit higher.
FAST METHOD ■	$12n > 180 \rightarrow n > 15 \rightarrow$ minimum integer $n = 16$ ✓

Q49. [HARD] ✓ **Correct Answer: A) V = 8,000 + 350y**

An investor puts $5,000 in Account A earning $200 per year (simple interest) and $3,000 in Account B earning $150 per year. After y years, which equation gives ...

STEP-BY-STEP SOLUTION:

Account A: 5,000 + 200y. Account B: 3,000 + 150y. Combined: (5,000 + 3,000) + (200 + 150)y = 8,000 + 350y.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: sum of both principals and both annual interest amounts.
✗ **B)** Swaps the interest rates — $150 goes with Account A and $200 with Account B. This gives V = 5,000 + 150y + 3,000 + 200y = 8,000 + 350y (happens to simplify to the same total, but the assignment is wrong).
✗ **C)** V = 350y ignores the $8,000 combined principal.
✗ **D)** V = 8,350y treats the total first-year value as a multiplicative rate — wrong structure.

TRAP ALERT ■	Failing to combine like terms — students sometimes forget to add both principal values together.
KEY INSIGHT ■	Combine: principals add, rates add. V = (P■ + P■) + (r■ + r■)y.
FAST METHOD ■	At y = 1: V = 8,000 + 350 = $8,350. Verify: A = $5,200; B = $3,150. Sum = $8,350 ✓

Q50. [HARD] ✓ **Correct Answer: A) S = 5t + 20**

A car accelerates from 20 mph to 65 mph over 9 seconds. Assuming constant acceleration, which linear equation models the speed S, in mph, after t seconds?

STEP-BY-STEP SOLUTION:

At t = 0: S = 20 mph. At t = 9: S = 65 mph. Slope = (65 – 20)/9 = 45/9 = 5. S = 5t + 20.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: starts at 20 mph, increases 5 mph per second.
✗ **B)** Slope of 45 uses total change, not change per second.
✗ **C)** Uses final speed as intercept — wrong.
✗ **D)** Uses 9 (time interval) as slope — wrong.

TRAP ALERT ■	Using the total speed change (45 mph) as the slope instead of dividing by 9 seconds.
KEY INSIGHT ■	Slope = total speed change / total time = 45/9 = 5 mph per second.
FAST METHOD ■	S(9) = 5(9) + 20 = 45 + 20 = 65 ✓

TOPIC SUMMARY — CREATING LINEAR MODELS FROM WORD PROBLEMS

Common Mistakes Students Make:

1. Swapping the slope (rate) and y-intercept (fixed fee) when writing the equation.
2. Using + instead of – for quantities that decrease over time (burning, draining, depreciating).
3. Forgetting to find the NET rate when two quantities are changing simultaneously (e.g., fill vs. drain).

Must Remember — Rules & Formulas:

1. $y = mx + b$: m = rate of change (per unit), b = starting/fixed value (when $x = 0$).
2. 'Already has' / 'starts with' / 'flat fee' = b (y-intercept).
3. Always verify your model by substituting the given values back in.

Expected on DSAT:

3-5 questions per DSAT test (appears in both modules). Consistently tested as a high-frequency topic.